Generation of Guide Star Catalog for Star Trackers

Jian Li, Gangyi Wang, and Xinguo Wei

***Abstract*— In this paper, a novel guide star catalog generation algorithm is presented, which is a crucial part of an advanced star tracker design since the performance and reliability of star identification and attitude determination depend on the guide star selection. First, we propose an analytical method to estimate the stellar instrument magnitude and characterize the associated errors. Those stars whose instrument magnitudes are higher than the star tracker sensitivity constitute the original catalog. Then, the probability model of each star in the original catalog for attitude determination is established as the function of the field of view (FOV), the brightness accuracy, and the mean number of stars in FOV based on the brightness and uniformity principles, and the stars with larger probability are selected as guide stars. The accuracy of photometry method is validated by estimating the Hipparcos catalog instrument magnitude and comparing them with actual data. The efficiency of guide star selection algorithm is demonstrated by generating guide star catalog for a star tracker with a 16° × 16° FOV.**

***Index Terms*— Star tracker, guide star catalog, stellar instrument magnitudes.**

I. Introduction

As a kind of high-precise attitude measurement device, star trackers are widely used in both orbiting and interplanetary spacecraft [1]. A star tracker uses observation unit vectors toward measured stars in the body frame and corresponding reference unit vectors of guide stars known in the inertial frame to generate three-axis attitude determinations [2].

As a basis for star identification and attitude determination, the guide star catalog, which is composed of right ascensions, declinations (or direction cosines) and instrument magnitudes of guide stars, is closely related to the star sensor's storage capacity, the execution speed of the star pattern identification algorithm and the design of the internal optical system. Hence, guide star catalog generation algorithm is a crucial part of an advanced star tracker design since the performance and reliability of star identification and attitude determination depend on the guide star selection [3], [4]. Obviously, it is necessary that guide stars could be imaged by the sensor. Further, guide stars should be selected such that they are relatively uniform across the celestial sphere to implement an unbiased identification and attitude determination routine. The less the guide stars in the FOV, the more difficult the star identification and the lower the accuracy of the attitude determination. on the other hand£ too many guide stars in the FOV contribute less to the identification rate and the attitude accuracy [1], [5], but increase the recognition and calculation time, as well as the memory requirement. Hence, the guide star catalog should contain as few stars as possible to achieve the desired recognition rate and attitude accuracy for better time and memory performance.

Existing guide star selection algorithms can be partitioned into two classes. The first class is magnitude filtering method (MFM), which selects stars brighter than or equal to a given magnitude threshold as guide stars. The magnitude threshold should he selected by considering starlight sensitivity of the star tracker. The most direct and common method takes the visual magnitude in some astrometric catalog as a criterion [6], which introduces a large error about 0.4 magnitude as the spectral response of a star tracker is not the same as the human eye. Hence, some methods are proposed to estimate the instrument magnitudes for stars [7]–[9]. Basically, the procedure involves the use of published detailed stellar spectral irradiance data [10] and sensor spectral responsivity to numerically determine the instrument magnitudes. For stars whose photometric data are not available, those above computed instrument magnitudes together with the generally available colour index are used to obtain the instrument magnitude through a polynomial regression. The estimation error is within 0.2 magnitude due to incomplete sample data and use of the same polynomial to describe different spectral types. In addition, Smith uses publicly available flight data from the three ICESat star trackers to predict instrument magnitudes which is not applicable to most star trackers as it is difficult to obtain the flight data [11]. Hence, the instrument magnitude estimation methods still need to be improved. Generally, the MFM is very easy to apply and provides well matching guide star set with measurements, but the distribution of guide stars varies significantly over the sky which biases the identification routine and seriously degrades the time and memory performance [2].

The second class of guide star selection algorithms is geometric filtering method (GFM), which selects stars such that such that guide stars are relatively uniform across the celestial sphere. Vedder [12], Silverman [13] and Bauer [14] use a set of uniformly distributed reference points on a celestial sphere.

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The star close to each reference point was selected as guide star. The methods could not provide all the information about holes. Hence, Kim and Junkins [15] presents a self-organizing guide star selection algorithm and Zhang *et al.* [16] further refines it by employing Boltzmann entropy to optimize the selection of multiple neighboring stars. The methods provide a well-balanced guide star catalog without possibility of holes and with near minimum number of guide stars, and guarantee at least two guide stars are contained in every possible boresight direction of the star tracker, which is not enough for the star identification rate as well as attitude determination accuracy. Basically, the GFM excludes some bright stars from the guide star catalog which would decrease the attitude accuracy as the brighter star is tracked more stably and measured more accurately.

To take into account the brightness and uniformity simultaneously, Li *et al.* [17] divides the celestial sphere into a number of equal-areas and contiguous cells, and then selects the brightest one or two stars in each cell as guide stars. It is shown that guide star catalog generated by the methods for a star tracker with an $12° \times 12°$ FOV achieves the requirements in terms of both functionality and performance. But Li does not discuss the selection of parameters such as the number of cells and stars in each cell, as well as their relation with FOV. Hence, the approach is not considered a guide star selection method for general purposes.

Based on the principles of brightness and uniformity, we propose an analytical method of generating guide star catalog. The method performs a complete search for the original star catalog rather than Monte Carlo simulation. Firstly, a method of estimating the stellar instrument magnitude is proposed, which improves the precision of the polynomial regression algorithm. Secondly, with the brightness error, the FOV and mean number of guide stars measured in the FOV as independent variables, the star selection probability model is established, and the star with larger probability is selected as the guide star. Typically, our method, quantifying the correlation between star tracker performance and guide star distribution, is suitable for general purposes.

The remainder of this paper is organized as follows. Section II describes an analytical method to estimate the stellar instrument magnitude and characterize the associated errors. Then, the original star catalog including stars brighter than or equal to a given instrument magnitude threshold is obtained. Section III establishes the guide star selection probability model, generates the guide star catalog, and presents the experiment results to demonstrate the performance of the proposed algorithm. Finally, some conclusions are drawn.

II. Stellar Instrument Magnitude Estimation

The magnitude indicates the relative brightness of the stars in the sky. By definition, the brightness of the star with magnitude difference of 5 varies by 100 times. The magnitude is also related to the spectral responsivity of the receiver. For example, the apparent magnitude is calculated by the spectral responsivity of the human eye, while the instrument magnitude is calculated according to the spectral responsivity of the imaging sensor.

The original star catalog is obtained by selecting stars brighter than or equal to a given instrument magnitude threshold. The star catalog quality and the star tracker performance are affected by the star tracker sensitivity and the instrument magnitude accuracy.

Herein an original star catalog is simulated. The star sensor configuration makes use of a 20 degrees circular FOV. The star tracker sensitivity is set to 5.4Mv instrument magnitude so that the number of the visible stars is 2405. The estimation deviation of instrument magnitude is 0.2 Mv.

When the instrument magnitude threshold is set to 5.4 Mv, then the original star catalog contains 2468 stars. By comparison, 265 visible stars are lost in the original star catalog, while 328 invisible stars are added. The lost stars could reduce full-sky star recognition rate and attitude accuracy.

Basically, the original catalog should contain all of the visible stars that can actually be imaged by a star tracker. The instrument magnitude threshold is set to 6.0 Mv so that the original star catalog contains 4999 stars. By comparison, the original star catalog contains 2594 invisible stars, which degrades the memory and time performance of star tracker.

Therefore, the more accurate the instrument magnitude, the lower the magnitude threshold, and the smaller the original star catalog capacity.

In this section, an analytical method is proposed to estimate the stellar instrument magnitude, and get the original star catalog.

A. Method Description

Initially, the relationship between the stellar color index and its temperature is derived by approximating the relative spectral characteristics of the stars to blackbody radiation. Then the star temperature is calculated by use of the color index $M_{\mathrm{B}} - M_{\mathrm{V}}$ and the spectral responsivity of corresponding detectors presented in the Tycho catalog to obtain the relative spectral characteristics of the stars. Finally, the stellar instrument magnitude is estimated from the definition.

The relative radiation from a star at a given wavelength and temperature is the same as a blackbody radiator and given by [18],

$$\mathrm{i}\,(\lambda, \mathrm{T}) = \frac{2\pi hc^2}{\lambda^5\left(\mathrm{e}^{hc/\lambda k_{\mathrm{B}}T} - 1\right)} \tag{1}$$

Where h is Planck constant $= 6.626 \times 10^{-34}\mathrm{J} \cdot \mathrm{s}$, c is the speed of light $= 2.997 \times 10^8\mathrm{m/s}$, and k_{B} is Boltzmann constant $= 1.38 \times 10^{-23}\mathrm{J/K}$, λ is the wavelength in meter and T is the temperature in kelvin. The relative radiation characteristics of the stars with the same temperature are the same, while the absolute radiation characteristic, denoted by $f\,(\lambda, \mathrm{T})$, is obtained by multiplying a constant C to the relative radiation characteristic.

Formally, the stellar magnitude M may be defined as follows

$$M = 2.5\lg\int_{\lambda_{\mathrm{L}}}^{\lambda_{\mathrm{U}}} f_0(\lambda, T_0)s(\lambda)d\lambda - 2.5\lg\int_{\lambda_{\mathrm{L}}}^{\lambda_{\mathrm{U}}} f(\lambda, T)s(\lambda)d\lambda \tag{2}$$

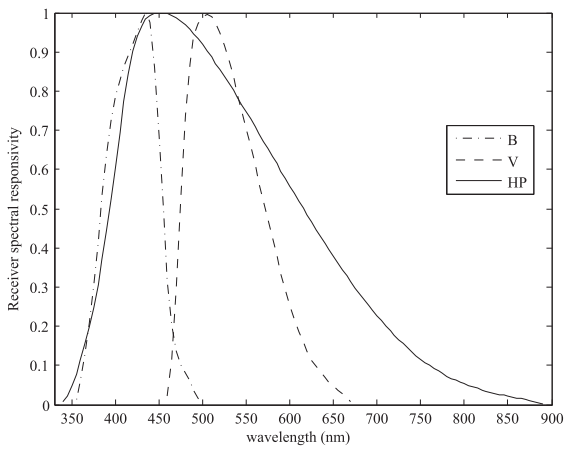

Fig. 1. Spectral responsivity of corresponding detectors presented in Tycho and Hipparcos catalog.

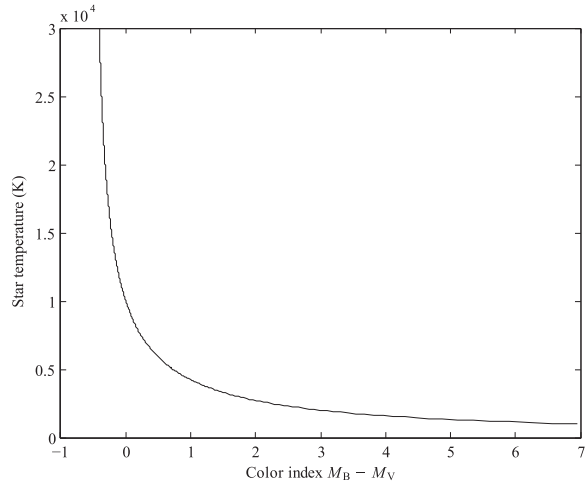

Fig. 2. Star temperature versus the color index $M_B - M_V$.

Where $f_0(\lambda, T_0)$ and $f(\lambda, T)$ are respectively the absolute radiation characteristics of 0 magnitude reference star of spectral class A0 and observed star, $s(\lambda)$ is spectral responsivity of corresponding detector, λ_L and λ_U are respectively the lower and upper limit of wavelength.

To facilitate analysis, two terms are introduced as follows

$$\mathrm{F}(\mathrm{T}) \equiv \int_{\lambda_L}^{\lambda_U} f(\lambda, \mathrm{T})\, \mathrm{s}(\lambda)\, \mathrm{d}\lambda \tag{3}$$

$$\mathrm{I}(\mathrm{T}) \equiv \int_{\lambda_L}^{\lambda_U} \mathrm{i}(\lambda, \mathrm{T})\, \mathrm{s}(\lambda)\, \mathrm{d}\lambda \tag{4}$$

Where $F(\mathrm{T})$ is different from $I(\mathrm{T})$ by the same scale factor C.

Color index is defined as the magnitude difference of the same star for detectors with different spectral responsivity.

$$\begin{aligned} &\mathrm{M}_1 - \mathrm{M}_2 \\ &= 2.5 \lg\left(\frac{\mathrm{F}_1(\mathrm{T}_0)}{\mathrm{F}_1(\mathrm{T})}\right) - 2.5 \lg\left(\frac{\mathrm{F}_2(\mathrm{T}_0)}{\mathrm{F}_2(\mathrm{T})}\right) \\ &= 2.5 \lg\left(\frac{\mathrm{F}_1(\mathrm{T}_0)}{\mathrm{F}_2(\mathrm{T}_0)} \cdot \frac{\mathrm{F}_2(\mathrm{T})}{\mathrm{F}_1(\mathrm{T})}\right) = 2.5 \lg\left(\frac{\mathrm{I}_1(\mathrm{T}_0)}{\mathrm{I}_2(\mathrm{T}_0)} \cdot \frac{\mathrm{I}_2(\mathrm{T})}{\mathrm{I}_1(\mathrm{T})}\right) \end{aligned} \tag{5}$$

Thus, the relationship between color index and star temperature is established from equation (5). Then the star temperature could be obtained given the blue magnitude M_B, apparent magnitude M_V and the spectral responsivity of corresponding detectors presented in the Tycho catalog as shown in Fig.1 [19]. Conversely, the star temperature and the spectral responsivity of star tracker could be used to calculate the color index $M_I - M_V$, as well as stellar instrument magnitude M_I.

As it is difficult to resolve an analytical expression of color index for the temperature, a numerical method is presented as follows. The integration interval (λ_L, λ_U) is divided into $2m$ parts with the same length $h = \frac{\lambda_U - \lambda_L}{2m}$, and equation (4) could be calculated according to Symposon's integration rule [20],

$$\begin{aligned} \mathrm{I} \approx \frac{h}{3}[&\mathrm{i}(\lambda_L)\, \mathrm{s}(\lambda_L) + \mathrm{i}(\lambda_U)\, \mathrm{s}(\lambda_U) \\ &+ 4\sum_{i=1}^{m} \mathrm{i}(\lambda_L + (2i-1)h)\, \mathrm{s}(\lambda_L + (2i-1)h) \\ &+ 2\sum_{i=1}^{m-1} \mathrm{i}(\lambda_L + 2ih)\, \mathrm{s}(\lambda_L + 2ih)] \end{aligned} \tag{6}$$

By substituting equation (1) and (6) into (5), the color index $M_B - M_V$ could be calculated for a given temperature as shown in Fig. 2. It can be seen that the color index increases monotonously with the decrease of temperature. Noted that the color index accuracy in the Tycho catalog reaches 0.01 magnitude, a lookup table is established to store the color index corresponding to each temperature at 50K intervals so that the difference between two adjacent color indices is less than 0.01 magnitude. Given the star color index, the nearest entry in the lookup table could be found to obtain the stellar surface temperature. Similarly, the instrumental magnitude of the star is obtained by calculating the color index between the instrumental magnitude and the apparent magnitude according to equation (5).

B. Experiment and Analysis

The photometry method is validated by generating instrument magnitudes (referred as Hp) in Hipparcos satellite photometry system and comparing them with the Hipparcos catalog instrument magnitudes with a precision of 0.001 magnitude. The instrument magnitudes of 25848 stars with apparent magnitudes ranging down to 7.5 based on the Hipparcos catalog are calculated. Fig. 3 shows the distribution of Hp estimation error. The root mean square error is 0.036 magnitude and the maximum error is about 0.2 magnitude.

In addition, the proposed method is compared with previous method in [7]. Firstly, we remove the stars that are not suitable for samples, such as variable stars, binary stars and

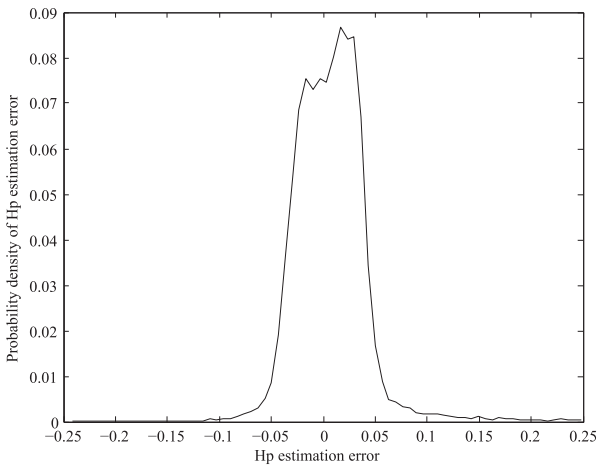

Fig. 3. Hp estimation error distribution of the proposed method. The statistics include 25848 stars with apparent magnitudes ranging down to 7.5 in the Hipparcos catalog.

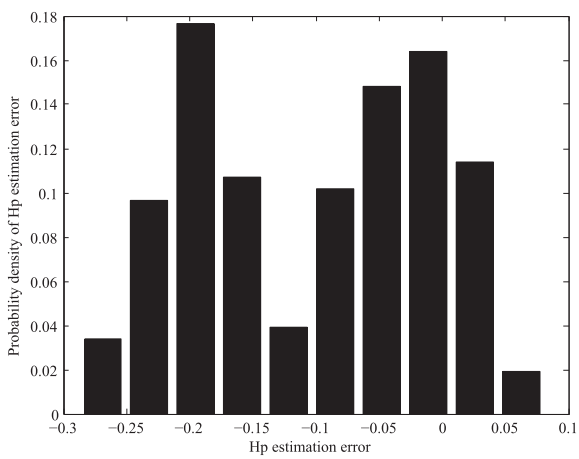

Fig. 4. Hp estimation error distribution of the reference star set using the 13-color photometry.

stars with incomplete spectral irradiance data, and reduce Johnson/Mitchel source star set [10] from 1380 stars to 754 stars termed the reference star set. Then the 13-color photometry is used to calculate Hp of the reference star set via a numerical solution of equation (2). Form equation (2), the integration over the wavelength is needed for the calculation of stellar magnitude. The 13-color photometry method approximates the integration by summation at 13 wavelengths. The wavelengths for the 13-color are not evenly distributed. The intervals between shorter wavelengths are smaller than that between longer wavelengths. For the higher temperature stars whose radiation characteristics change more greatly at shorter wavelengths and smaller at longer wavelengths, the Hp estimation of the 13-color photometry method is close to the true value. Conversely, for the lower temperature stars whose radiation characteristics change smaller at shorter wavelengths and more greatly at longer wavelengths, the Hp estimation of the 13-color photometry method is far from the true value. Hence, the error of the 13-color photometry method is biased. The biased error distribution is shown in Fig. 4. It can be seen that he mean error is about -0.1 magnitude and the root mean square error is 0.14 magnitude.

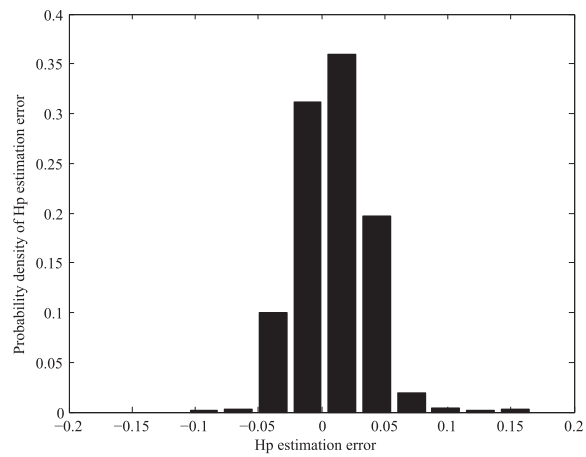

Fig. 5. Hp estimation error distribution of the reference star set using the proposed method.

The proposed method calculates the star temperature to obtain the relative spectral characteristics of the stars. And in the process of the stellar magnitude calculation the integration interval is equally divided with 5 micrometers space. So the proposed method is unbiased and more accurate than the 13-color photometry method. Fig.5 shows the distribution of Hp estimation error of the reference star set for our method. The error distribution is unbiased and the root mean square error is 0.029 magnitude which is decreased by 79%.

With the Hp obtained using the 13-color photometry above, we tabulate the color indices $M_{\mathrm{B}} - M_{\mathrm{V}}$ and $Hp - M_{\mathrm{V}}$ for stars in the reference star set. The M_{B} and M_{V} magnitudes are obtained from the corresponding star records in Tycho catalog. We form a linear regression on $Hp - M_{\mathrm{V}}$ versus $M_{\mathrm{B}} - M_{\mathrm{V}}$ for stars in the reference set. Hence for those stars which are not in the reference star set, Hp is obtained from the relation

$$Hp = 0.20474 M_{\mathrm{B}} + 0.79526 M_{\mathrm{V}} + 0.011448 \tag{7}$$

Fig. 6 shows the distribution of Hp estimation error of stars out of the reference star set. The root mean square error is 0.19 magnitude. In comparison, the precision of Hp estimated by our method is increased by 81%. Therefore, the stellar instrument estimation method proposed in this paper is accurate and feasible.

III. Guide Star Selection

Generally, there are two criteria to select stars for attitude determination. Firstly, brighter stars are preferred because the brighter stars have higher location accuracy. Secondly, the distribution of stars in the field of view (FOV) is as uniform as possible. This not only reduces the capacity of guide star database, but also improves the accuracy of attitude determination, as the uneven distribution results in biased estimation of the attitude angle cross boresight and reduces the estimation accuracy of the attitude angle around boresight.

The priority of a star for attitude determination changes with the FOV. In case there are fewer stars in FOV, or the stars in the neighborhood are darker, the star has a higher priority. Conversely, if there are more stars in FOV, or the stars in the

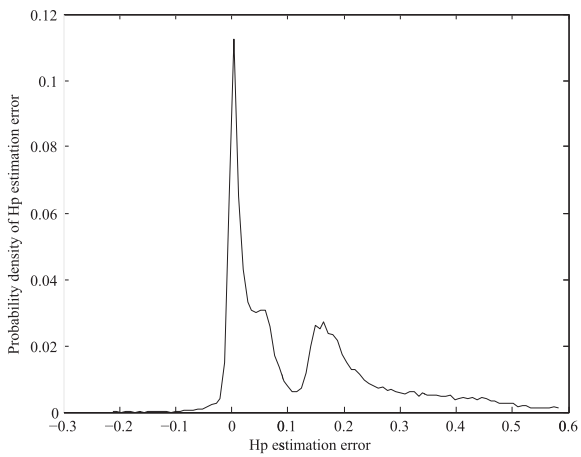

Fig. 6. Hp estimation error distribution of the stars which are not in the reference star set using the linear regression method.

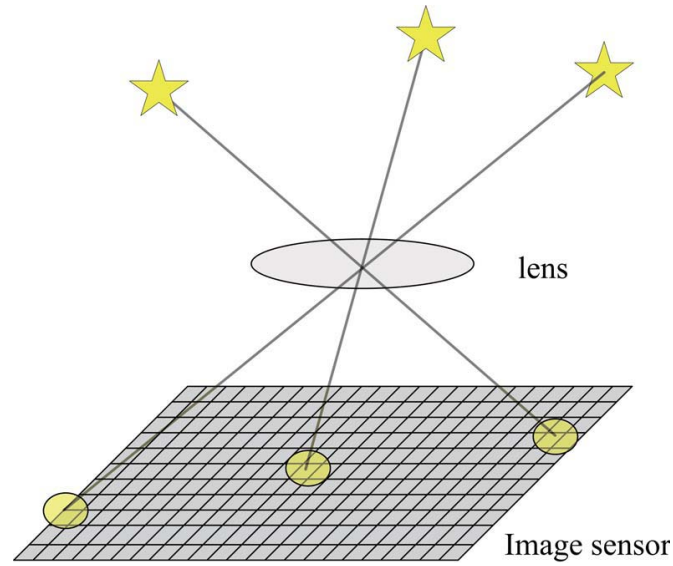

Fig. 7. Imaging modeling of a star tracker.

neighborhood are brighter, the star has a lower priority. Hence, all cases need be traversed for each star to determine whether it is selected as a guide star.

To avoid excessive computation, kim specifies that the number of stars used for attitude determination is not less than two, which is far from meeting the requirements of autonomous star identification and attitude accuracy. However, simply dividing the celestial sphere and then analyzing each region separately, in essence, is a trial-and-error method. The result is related to the dividing method, the number of sub-blocks and the design parameters of the star sensor which need to be improved according to the Monte Carlo simulation. To overcoming the above shortcoming, we analyze the characteristics of star tracker image, discretize the boresight distribution, establish the probability model of each star in the original catalog for attitude determination as the function of the FOV, the brightness accuracy, and the mean number of stars in FOV based on the brightness and uniformity principles, and select the stars with larger probability as guide stars.

A. Probability Model of Guide Star Selection

According to the conditional probability in probability theory, the probability that a star is used for attitude determination is equal to the sum of the product of the star selected probability in every direction and the boresight pointing probability.

$$P_{\mathrm{s}} = \iint\limits_{\theta(i,b)<(FOV/2)} P_{\mathrm{s}}|_{\mathrm{b}} \rho_{\mathrm{b}} d\mathrm{s} \tag{8}$$

Where $P_{\mathrm{s}}|_{\mathrm{b}}$ is star selected probability in some direction, ρ_{b} is the boresight pointing probability density function. The integral range is the area on the celestial sphere that is less than half of the FOV from the star, as the star could not enter the FOV when its angle distance from the boresight is greater than half of the FOV so that its condition probability is 0. It is assumed that the boresight follows a uniform distribution across the celestial sphere. The probability density ρ_{b} is thus given by [2]

$$\rho_{\mathrm{b}} = \frac{1}{4\pi} \tag{9}$$

Taking into account the actual use of the star tracker, the integral in equation (8) can be replaced by summation. The reasons are as follows: The star tracker optical system is usually aligned in a defocused position so that the size of the star image is 3×3 to 5×5 pixels to improve the star location measurement accuracy [21], as shown in Fig. 7. With a pixel resolution of N_{pixel}, the star tracker needs to rotate $3 \times \mathrm{FOV}/N_{\mathrm{pixel}}$ to $5 \times \mathrm{FOV}/N_{\mathrm{pixel}}$ perpendicular to the boresight to ensure that a star completely enters the FOV.

Therefore, the neighborhood on the celestial sphere that is less than $\mathrm{FOV}/2$ from a star is equally divided into $N_{\mathrm{boresight}}$ regions in two directions perpendicular to each other with the interval equal to $3 \times \mathrm{FOV}/N_{\mathrm{pixel}}$, as shown in Fig.8. And the area of a region is expressed as [1]

$$\Delta \mathrm{s} = \frac{2\pi(1 - \cos\frac{\mathrm{FOV}}{2})}{\mathrm{N}_{\mathrm{boresight}}} \tag{10}$$

Thus, the probability of a boresight falling in each region is

$$P(m,n) = \rho_{\mathrm{b}} \times \Delta \mathrm{s} = \frac{(1 - \cos\frac{\mathrm{FOV}}{2})}{2N_{boresight}} \tag{11}$$

The probability that a star will be selected for attitude determination in each region is donated by $P_{\mathrm{s}}|_{(m,n)}$, and then we have

$$P_{\mathrm{s}} = \sum P_{\mathrm{s}}|_{(m,n)} P(m,n) = \frac{(1 - \cos\frac{\mathrm{FOV}}{2})}{2N_{boresight}} \sum P_{\mathrm{s}}|_{(m,n)} \tag{12}$$

$P_{\mathrm{s}}|_{m,n}$ is calculated below. Taking into account the brightness and uniformity simultaneously, the stars in some FOV are selected for attitude determination as follows: For the remaining stars in the field of view, the angular distances between each star pair are calculated. Then the smallest angular distance is found and the darker one of the corresponding

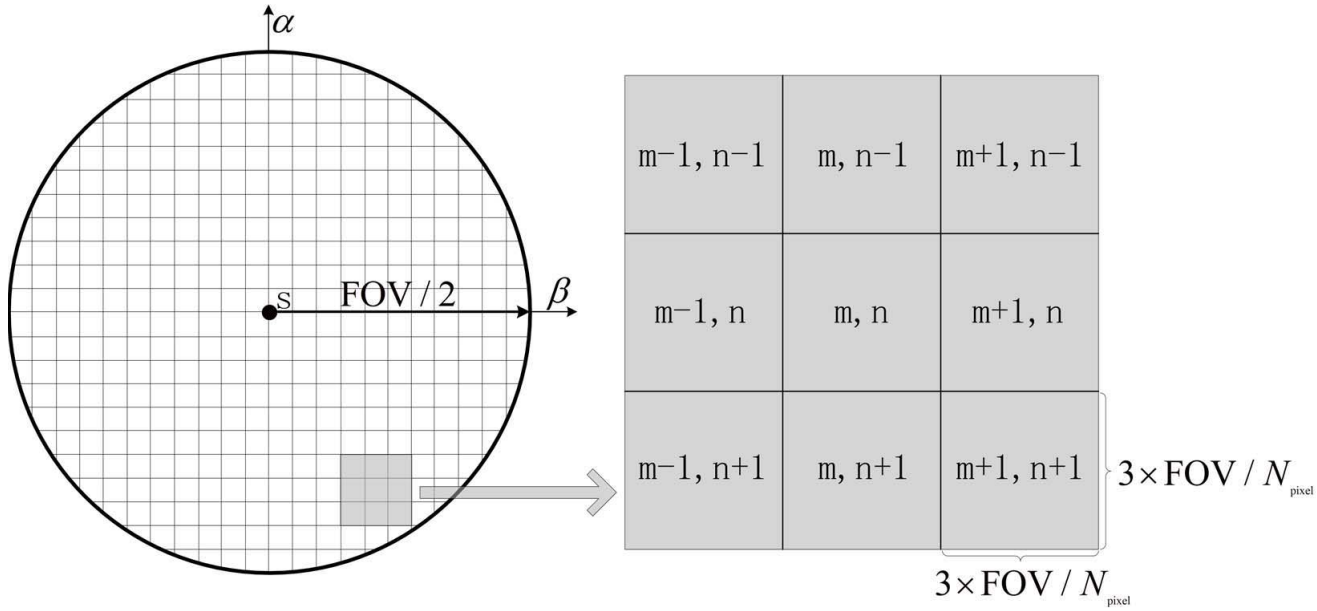

Fig. 8. Discretization of the boresight distribution. s respresents the target star, α and β represent the two directions perpendicular to each other respectively.

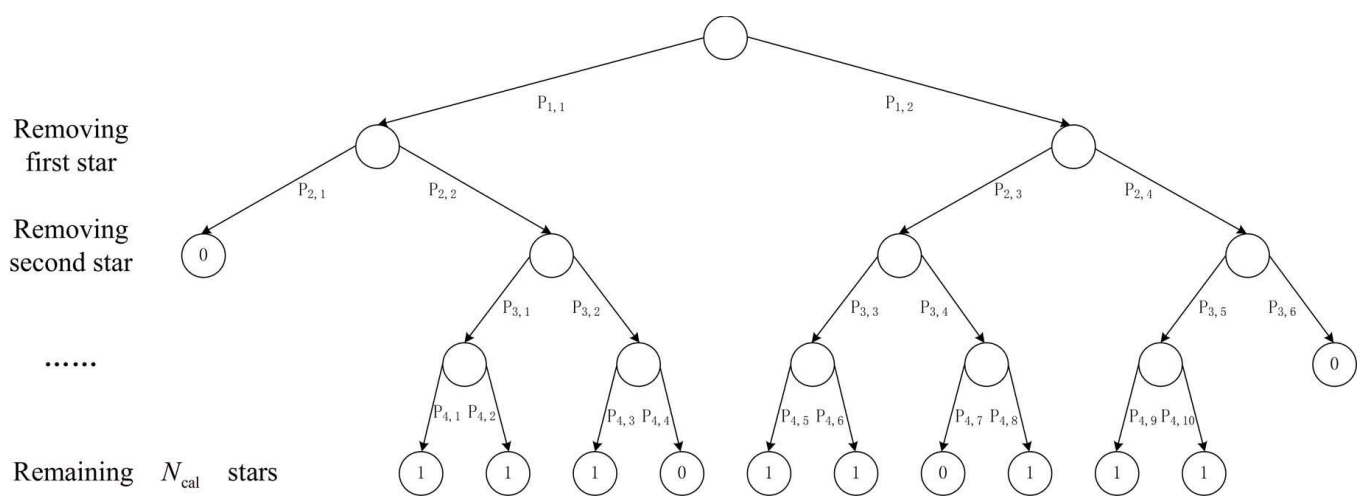

Fig. 9. Process of selecting the stars in the same FOV for attitude determination. There are 2 cases for each star reduction and $m = 2^{(n-N_{\text{cal}})}$ cases for selecting N_{cal} from n observed stars. The node marked 0 represents that the target star is removed, and the node marked 1 represents that the target star is remained.

two stars is removed until the number of remaining stars less than the upper limit of the number of guide stars used to calculate the attitude which depends on the processing capability and accuracy specification of the star tracker and is donated by N_{cal}.

For the star pair (i, j) with the smallest angular distance, the probability of removing i and j is denoted by $P_{\text{s}}(i)|_{(m,n)}$ and $P_{\text{s}}(j)|_{(m,n)}$ separately, obviously we have

$$P_{\text{s}}(j)|_{(m,n)} = 1 - P_{\text{s}}(i)|_{(\text{m,n})} \tag{13}$$

It is assumed that the brightness uncertainty of a star image, resulting from estimation residuals, optical system performance, imaging sensor noise, is Gaussian noise with zero mean and standard deviation σ_M, and the instrument magnitude of star i and j are M_i and M_j separately. Then the probability that star i is brighter than j is obtaining as

$$\begin{aligned} &P_{\text{s}}(i)|_{(m,n)} \\ &= P(M_i < M_j) = \frac{1}{2\pi\sigma_M^2} \\ &\quad \times \int_{-\infty}^{+\infty} \exp\left(-\frac{(\text{x} - M_i)^2}{2\sigma_M^2}\right) \int_{x}^{+\infty} \exp\left(-\frac{(y - M_j)^2}{2\sigma_M^2}\right) dy dx \end{aligned} \tag{14}$$

As shown in Fig. 9, there are 2 cases for each star reduction and $m = 2^{(n-N_{\text{cal}})}$ cases for selecting N_{cal} from n observed stars which could be divided into three classes.

In the first class, the two stars that correspond to the smallest angular distance are not the target star. As shown in Fig.9, one star is removed with probability $P_{1,1}$, and the other

TABLE I

CHANCE THAT THE NUMBER OF STARS IS LESS THAN SOME SPECIFIED VALUES AND THE STATISTICS OF THE NUMBER OF STARS IN DIFFERENT FOVS

probability threshold	Chance of Small Number of Measured Stars				Minimum number of stars	Maximum number of stars	Average number of stars	Standard deviation	Catalog capacity
	$N<5$	$N<6$	$N<7$	$N<8$					
1.00	0.49389	0.61596	0.72822	0.81910	0	15	4.63	2.93	951
0.95	0.00401	0.01188	0.03228	0.07512	1	21	11.13	2.45	2286
0.90	0.00048	0.00214	0.00881	0.02491	3	21	12.28	2.40	2522
0.85	0.00005	0.00053	0.00217	0.00934	4	22	12.94	2.30	2658
0.80	0.00000	0.00019	0.00082	0.00397	5	22	13.52	2.32	2778
0.75	0.00000	0.00015	0.00059	0.00290	5	25	14.10	2.32	2897
0.70	0.00000	0.00001	0.00016	0.00128	5	26	14.61	2.28	3002
0.65	0.00000	0.00001	0.00009	0.00073	5	26	15.07	2.26	3096
0.60	0.00000	0.00001	0.00009	0.00068	5	26	15.54	2.30	3192
0.55	0.00000	0.00001	0.00009	0.00067	5	27	15.95	2.39	3276
0.50	0.00000	0.00001	0.00009	0.00067	5	28	16.37	2.50	3363
0.45	0.00000	0.00001	0.00009	0.00066	5	28	16.77	2.63	3445
0.40	0.00000	0.00001	0.00009	0.00065	5	29	17.24	2.82	3541
0.35	0.00000	0.00001	0.00008	0.00058	5	29	17.59	2.99	3612
0.30	0.00000	0.00001	0.00008	0.00058	5	30	17.98	3.16	3693
0.25	0.00000	0.00001	0.00008	0.00058	5	31	18.45	3.43	3789
0.20	0.00000	0.00001	0.00008	0.00058	5	34	18.89	3.70	3880
0.15	0.00000	0.00001	0.00008	0.00058	5	35	19.44	4.15	3994
0.10	0.00000	0.00001	0.00008	0.00058	5	38	20.31	4.92	4173
0.05	0.00000	0.00001	0.00008	0.00058	5	45	21.52	6.35	4421
0.00	0.00000	0.00001	0.00008	0.00058	5	76	23.54	9.84	4837

star is removed with probability $P_{1,2}$. Regardless of the star which is removed, the number of observed stars is still greater than N_{cal}. In the subsequent selecting process, two situations need to be discussed separately.

In the second class, the two stars with the smallest angular distance contain the target star. The target star is removed with probability $P_{2,1}$, and the selecting process is terminated. The termination node is marked with 0 as shown in Fig.9. The target star is remained with probability $P_{2,2}$, and the number of observation stars is still greater than N_{cal} so that selecting process needs to be continued at this time.

In the third class, the two stars that correspond to the smallest angular distance are not the target star. After removing one star, the number of observed stars is equal to N_{cal}, and the selecting process is terminated. As shown in the last line of Fig.9, the target star is used for attitude calculation and the termination node is marked with 1 at this time.

The probability of each case is the product of all probabilities from the initial node to the terminated node. There are two termination nodes. For the node marked 0 in the figure, the number of remaining stars in the image is greater than or equal to N_{cal}, but the star is to be removed, i.e., the probability of the star for attitude determination is 0. For the node marked 1, the number of remaining stars in the image is equal to N_{cal}, and the star is used for attitude determination, i.e., the probability is 1. The probability that a star is used in the attitude determination for a given boresight direction is the sum of the probabilities that the cases with the terminated node marked 1 occur.

The stars whose probabilities are greater than a certain threshold are selected as guide stars. For an example, the threshold could be selected so that the expectation number of guide stars in the FOV is equal to N_{cal}.

B. Performance Verification of Guide Star Catalog

The guide star selection algorithm presented in this paper is evaluated by simulation. The star tracker configuration used for the simulations reported in this paper uses a 16° circular FOV with a focal plane consisting of 2048 × 2048 pixels so that the region size shown in Fig. 8 is set to 0.023°. The original star catalog is made of 4837 stars below magnitude 6, based on Hipparcos catalog. The upper limit of the number of guide stars used to calculate the attitude, N_{cal}, is set to 16 according to the current star sensor technical specifications [22]–[25]. The brightness accuracy is set to 0.2 magnitude.

The probability that each star is used for attitude determination is calculated as described above. Then, the relationship between star selection probability and star distribution is analyzed. According to the method described in [26], 906012 uniformly distributed boresight directions on the celestial sphere are generated at an interval of 0.24°. Table I shows the chance that the number of stars, denoted by N, is less than some specified values tested in the 906012 boresight directions, as well as the statistics of the number of stars in different FOVs for different normalized probability threshold.

From the table some useful conclusions are drawn below. First of all, as seen in the dark gray row, the stars with probabilities greater than 0.55 are selected as guide stars so that the average number of stars in FOV is about N_{cal}. The guide star catalog contains 3276 guide stars which is decreased by 32.3% compared to the original catalog marked in the light gray, while the minimum number of stars in FOV is 5 which is equal to that of original catalog. The guide star selection method described in this paper retains the stars

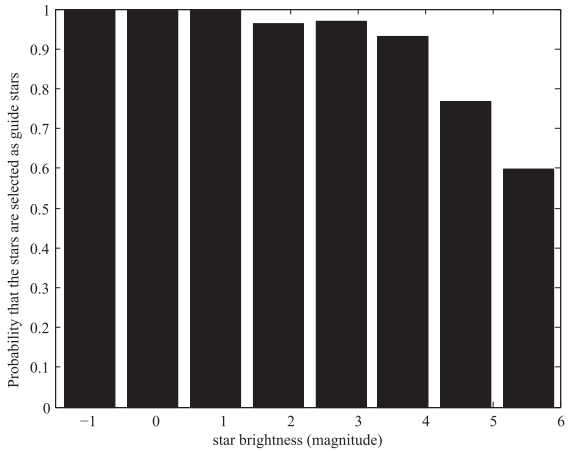

Fig. 10. Probability statistics that the stars of different brightness are selected as guide stars.

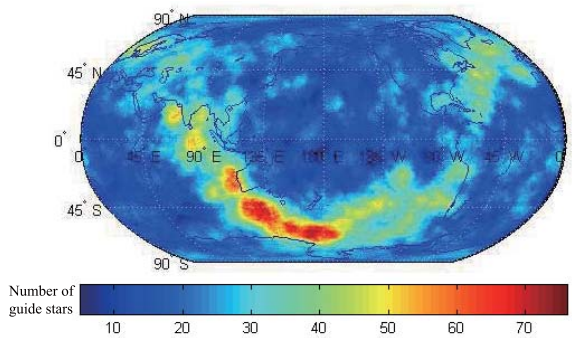

Fig. 11. Star density of the original star catalog before selection. Different colors represent different numbers of stars in FOV for a given boresight direction.

in the sparsely distributed regions over the celestial sphere. There are 27 stars at most in the FOV, which is greatly improved compared with the maximum of 76 stars for the original catalog. Fig.10 shows the probability statistics that the stars of different brightness are selected as guide stars. It is seen that brighter stars are retained to meet the brightness priority principle. Fig.11 and Fig.12 show the star distribution before and after the selection respectively in color codes. Different colors represent different numbers of stars in FOV for a given boresight direction. The standard deviation using the proposed method is 2.39 and is decreased by 76% compared to that of the original catalog, which shows that the uniformity of guide star catalog after selection greatly improves.

Secondly, as shown in Fig.13 to Fig.15, when the probability threshold decreases from 1 to 0.8, the capacity of the guide star catalog and the average number of guide stars in FOV increase rapidly, while the standard deviation decreases which means that the uniformity of the guide star catalog is improving. When the probability threshold decreases from 0.8 to 0.3, the guide star catalog capacity and the average number of guide stars in FOV are both linearly related to the threshold, and the standard deviation is always relatively small at about 2.5. In the process of the probability threshold decreasing from 0.3 to 0, the capacity of the guide star catalog

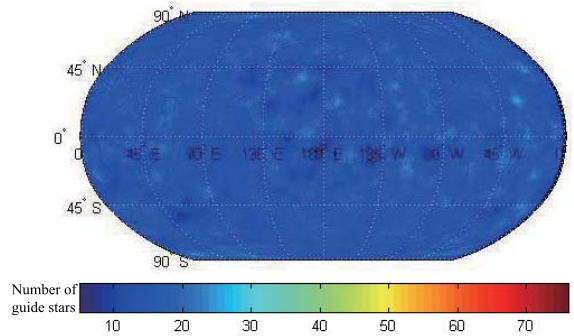

Fig. 12. Star density of the guide star catalog after selection.

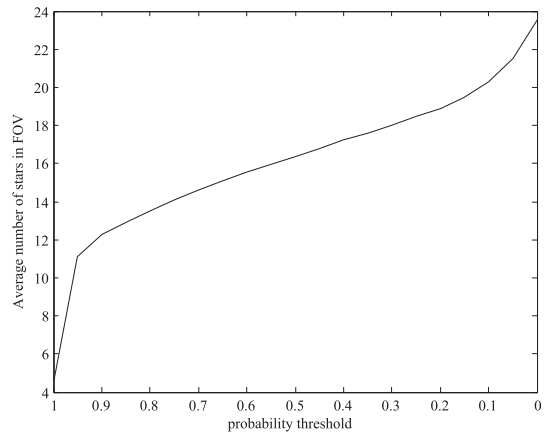

Fig. 13. Average number of stars in FOV versus probability threshold.

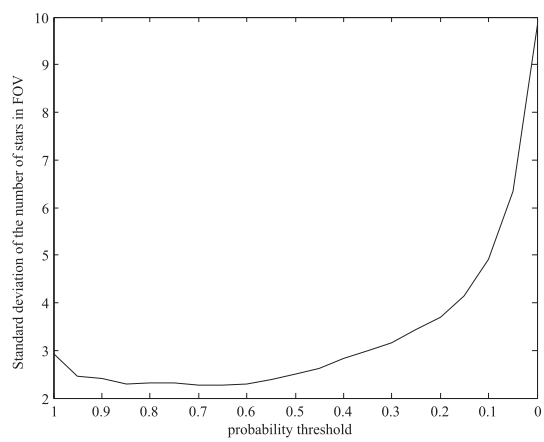

Fig. 14. Standard deviation of the number of stars in FOV versus probability threshold.

and the average number of guide stars in FOV increase rapidly, as well as the standard deviation indicating the degradation of the guide star catalog uniformity. Therefore, the probability threshold could be selected in the linear range from 0.3 to 0.8 so that the capacity of the guide star catalog and the average number of guide stars in FOV could meet the requirement of the memory and accuracy. For an example, if the average number of guide stars in FOV is 14 in the star tracker design, the threshold could be set to 0.75. In case that the average number of guide stars in FOV is

TABLE II
COMPARISON OF GUIDE STAR CATALOG PERFORMANCE BETWEEN THE PROPOSED METHOD AND THE SPACE SOLID ANGLE METHOD

	Chance of Small Number of Measured Stars				Minimum number of stars	Maximum number of stars	Average number of stars	Standard deviation	Catalog capacity
	$N<5$	$N<6$	$N<7$	$N<8$					
Our method	0.00000	0.00019	0.00082	0.00397	5	22	13.52	2.32	2778
The space solid angle method	0.00010	0.00108	0.00524	0.01835	3	30	13.89	3.58	2853

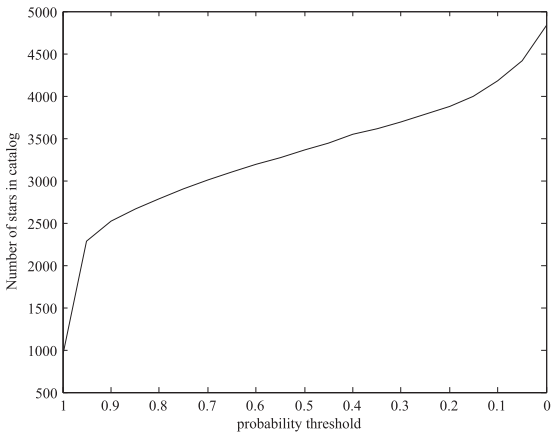

Fig. 15. Catalog capacity versus probability threshold.

not in the linear range, the probability of each star should be recalculated.

To illustrate the superiority of the proposed method, the comparison with the space solid angle method in [15] is performed under the premise that the catalog capacity and the average number of stars are basically the same. As shown in Table II, the probabilities of less stars in FOV is lower and the minimum number of stars in FOV is larger for the new algorithm. In addition, the maximum number of stars is 22 which is greatly improved compared with the maximum of 30 stars for the space solid angle method. The standard deviation gained through the proposed method is 64.8% of that gained through the space solid angle method, which shows the improvement of guide star catalog uniformity.

IV. CONCLUSION

In this study, a novel guide star catalog generation algorithm is presented for reliable and efficient performance of the star tracker. Obviously, this process is constrained such that the stars brighter than some sensor-dictated magnitude threshold are selected for the original star catalog. Hence, a method of estimating the stellar instrument magnitude is proposed firstly. The key of the method lies in the calculation of star temperature by use of the color index $M_{\mathrm{B}} - M_{\mathrm{V}}$ and the spectral responsivity of corresponding detectors presented in the Tycho catalog. By estimating the Hipparcos catalog instrument magnitude and comparing them with actual data, it was found that the root-mean-square deviation of error is 0.036 magnitude which is decreased by nearly 80% than that of polynomial regression method. Therefore, the stellar instrument estimation method proposed in this paper is accurate and feasible.

Secondly, the star selection probability is calculated, and the star with larger probability is selected as the guide star. This guide star selection approach avoids both the high dimensional optimization problem and the requirement of the Monte Carlo simulation for each configuration. Through the probability model obtained by quantifying the brightness and uniformity criterion, the relationship between the FOV, the brightness accuracy, the mean number of stars in FOV and the distribution of guide stars is established. Once the technical specifications of the star tracker are determined, the corresponding guide star catlog is obtained. Given the assumption that the star sensor brightness noise is 0.2 magitude, the FOV is 16° and the mean number of stars in FOV is 16, the simulation shows that the capacity of the selected guide star catalog is reduced by 32.3%, the brighter stars are retained and the distribution of guide stars is more even with the standard deviation of 2.39 which is decreased by 76% compared to that of the original catalog.

In summary, the method of guide star catalog generation described in this paper provides an analytical solution in the star tracker design for general purposes, which allows continuous operation of the star tracker and better performance.

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